

Chapter 3

Laplace Transform

Learning Objectives

By the end of this chapter, students should be able to

1. evaluate Laplace and inverse Laplace transforms of elementary functions;
2. evaluate Laplace and inverse Laplace transforms of a product involving e^{at} ;
3. evaluate Laplace and inverse Laplace transforms of a truncated shift of a function;
4. solve initial value problems (IVP) using Laplace transforms.

Laplace transforms derive their name from the French scholar **Pierre-Simon Laplace** (1749 – 1827), who did pivotal work in mathematics, statistics, physics, and astronomy. These transforms are useful for solving engineering problems involving initial value problems (IVP).

Section 3.1 introduces the definition of the Laplace transform and its basic properties. This is followed by a discussion on the Laplace transforms of elementary functions (polynomials, exponential, and trigonometric functions), as well as the unit-step function, the delta function, and convolution of functions.

Sections 3.2 and 3.3 are dedicated to evaluating Laplace transforms of products involving exponential function e^{at} or unit-step function $U(t - a)$; whereas Section 3.4 presents the Laplace transform as an elegant algebraic tool for solving IVPs of linear differential equations. As usual, we conclude the chapter with exercises in Section 3.5.

3.1 Definition and Formulas

In engineering applications, we commonly use a mathematical function in time (or a function in t -domain) to represent a signal, for example, electrical signal. The Laplace transform maps input functions in t -domain to output functions in s -domain (see Figure 3.1 and Definition 3.1.1).

We use lower-case letters to denote functions in the t -domain, and upper-case letters to denote functions in the s -domain. Figure 3.1 illustrates the role of Laplace transform.

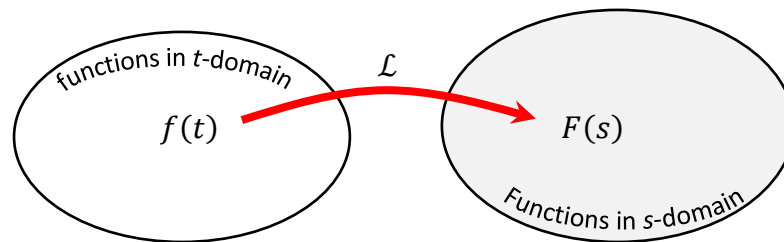


Figure 3.1. The Laplace transform maps functions in the t -domain to functions in the s -domain

Definition 3.1.1 (Laplace Transform)

The **Laplace transform** of $f(t)$, is a function $F(s)$, defined by

$$F(s) = \lim_{R \rightarrow \infty} \int_0^R e^{-st} f(t) dt$$

We use the notation $\mathcal{L}[f(t)]$ to denote the Laplace transform of $f(t)$.

Remark. The notation $\lim_{R \rightarrow \infty} \int_0^R e^{-st} f(t) dt$ above can be written as $\int_0^\infty e^{-st} f(t) dt$, i.e. as an *improper integral*.

The inverse Laplace transform has the opposite role, as illustrated by Figure 3.2.

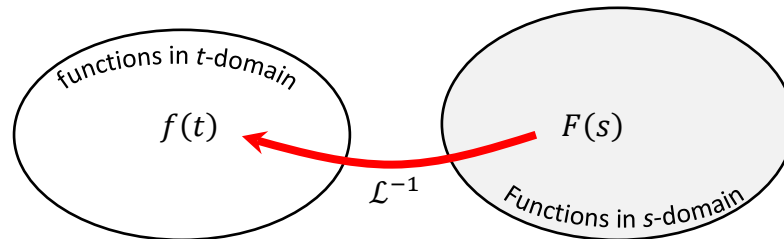


Figure 3.2. The inverse Laplace transform maps functions in the s -domain to functions in the t -domains

The Laplace transforms of elementary functions are shown without proof in Table 3.1. We shall use them directly for subsequent problems.

$f(t)$	$\mathcal{L}[f(t)]$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin at$	$\frac{a}{s^2 + a^2}$
$\cos at$	$\frac{s}{s^2 + a^2}$
$\mathcal{L}^{-1}[F(s)]$	$F(s)$

Table 3.1 Laplace transforms of elementary functions

Remark. For $n = 0, 1, 2, \dots$, the notation $n!$ is read n factorial. It is the product of all integers from 1 to n , in other words, $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$.

The first two formulas in Table 3.1 can be written as follows.

$$\begin{array}{ll} \mathcal{L}[t^n] = \frac{n!}{s^{n+1}} & \text{or} \quad \mathcal{L}^{-1}\left[\frac{n!}{s^{n+1}}\right] = t^n \\ \mathcal{L}[e^{at}] = \frac{1}{s-a} & \text{or} \quad \mathcal{L}^{-1}\left[\frac{1}{s-a}\right] = e^{at} \end{array}$$

Example 3.1.2

- (a) $\mathcal{L}[t^3] = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$
- (b) $\mathcal{L}[t^4] = \frac{4!}{s^{4+1}} = \frac{24}{s^5}$
- (c) $\mathcal{L}^{-1}\left[\frac{6}{s^4}\right] = \mathcal{L}^{-1}\left[\frac{3!}{s^{3+1}}\right] = t^3$
- (d) $\mathcal{L}^{-1}\left[\frac{1}{s^3}\right] = \frac{1}{2!}\mathcal{L}^{-1}\left[\frac{2!}{s^{2+1}}\right] = \frac{1}{2}t^2$

Example 3.1.3

- (a) $\mathcal{L}[e^t] = \frac{1}{s-1}$
- (b) $\mathcal{L}[e^{-t}] = \frac{1}{s-(-1)} = \frac{1}{s+1}$
- (c) $\mathcal{L}^{-1}\left[\frac{1}{s-1}\right] = e^t$
- (d) $\mathcal{L}^{-1}\left[\frac{1}{s+2}\right] = \mathcal{L}^{-1}\left[\frac{1}{s-(-2)}\right] = e^{-2t}$

The Laplace transform for sine and cosine functions can be written as follows.

$$\mathcal{L}[\sin at] = \frac{a}{s^2 + a^2} \quad \text{or} \quad \mathcal{L}^{-1}\left[\frac{a}{s^2 + a^2}\right] = \sin at$$

$$\mathcal{L}[\cos at] = \frac{s}{s^2 + a^2} \quad \text{or} \quad \mathcal{L}^{-1}\left[\frac{s}{s^2 + a^2}\right] = \cos at$$

Example 3.1.4

$$(a) \quad \mathcal{L}[\sin 2t] = \frac{2}{s^2 + 2^2} = \frac{2}{s^2 + 4}$$

$$(b) \quad \mathcal{L}[\cos 2t] = \frac{s}{s^2 + 2^2} = \frac{s}{s^2 + 4}$$

$$(c) \quad \mathcal{L}^{-1}\left[\frac{1}{s^2 + 4}\right] = \frac{1}{2} \mathcal{L}^{-1}\left[\frac{2}{s^2 + 2^2}\right] = \frac{1}{2} \sin 2t$$

$$(d) \quad \mathcal{L}^{-1}\left[\frac{s}{s^2 + 4}\right] = \mathcal{L}^{-1}\left[\frac{s}{s^2 + 2^2}\right] = \cos 2t$$

The **unit-step function**, or Heaviside function, is a piecewise defined function given by

$$U(t - a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t \geq a \end{cases}$$

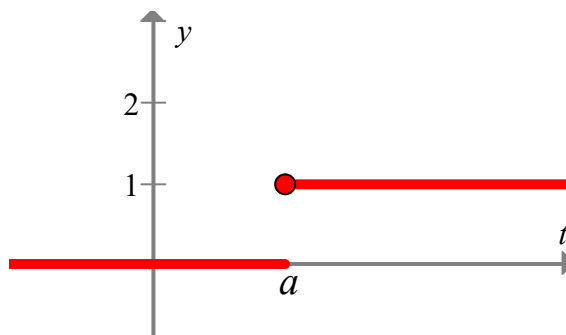


Figure 3.3. Graph of $U(t - a)$

Given a function $f(t)$, the product $f(t)U(t - a)$ describes the truncated function at $t = a$, that is, the function is cut-off prior to $t = a$ and resumes from $t = a$ onwards. Figure 3.4 shows the graphical comparisons between $f(t)$ and $f(t)U(t - a)$.

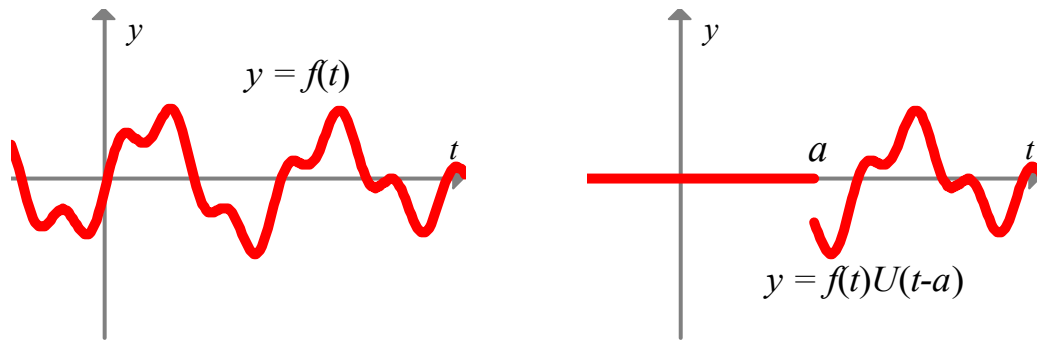


Figure 3.4. Graphs of $f(t)$ and $f(t)U(t - a)$

Another function that is very useful in engineering is called **Dirac's delta function**, $\delta(t - a)$, named after the British physicist, *Paul Adrien Maurice Dirac* (1902 – 1984). It is also known as the impulse function. Mathematically speaking, Dirac's delta function is not really a function in the formal sense. It satisfies the following two properties:

$$\delta(t - a) = \begin{cases} \infty & \text{for } t = a \\ 0 & \text{for } t \neq a \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1.$$

The delta function can be considered as the derivative of the unit-step function. The Laplace transform of the unit-step and delta functions is as follows.

$f(t)$	$\mathcal{L}[f(t)]$
$U(t - a)$ for $a \geq 0$	$\frac{e^{-as}}{s}$
$\delta(t - a)$ for $a \geq 0$	e^{-as}
$\mathcal{L}^{-1}[F(s)]$	$F(s)$

Table 3.2 Laplace transforms of unit-step and delta functions

The formulas in Table 3.2 can be written as follows.

$$\mathcal{L}[U(t - a)] = \frac{e^{-as}}{s} \quad \text{or} \quad \mathcal{L}^{-1}\left[\frac{e^{-as}}{s}\right] = U(t - a)$$

$$\mathcal{L}[\delta(t - a)] = e^{-as} \quad \text{or} \quad \mathcal{L}^{-1}[e^{-as}] = \delta(t - a)$$

Example 3.1.5

$$(a) \quad \mathcal{L}[U(t - 2)] = \frac{e^{-2s}}{s}$$

$$(c) \quad \mathcal{L}^{-1}\left[\frac{e^{-s}}{s}\right] = U(t - 1)$$

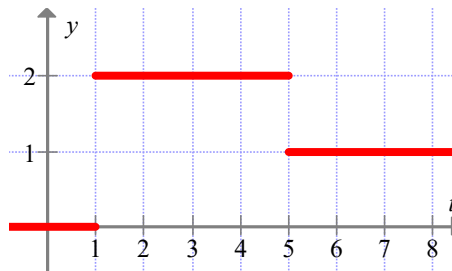
$$(b) \quad \mathcal{L}[\delta(t - 2)] = e^{-2s}$$

$$(d) \quad \mathcal{L}^{-1}[e^{-s}] = \delta(t - 1)$$

We can construct a step function by adding or subtracting unit-step functions. Example 3.1.6 shows how this is done.

Example 3.1.6

The graph of $f(t)$ is given below. Express $f(t)$ in terms of unit-step functions, and evaluate $\mathcal{L}[f(t)]$.



Solution

$$f(t) = 2U(t - 1) - U(t - 5)$$

$$\mathcal{L}[f(t)] = 2\mathcal{L}[U(t - 1)] - \mathcal{L}[U(t - 5)] = \frac{2e^{-s}}{s} - \frac{e^{-5s}}{s}$$

Laplace transforms and inverse Laplace transforms take on the linear properties of the integral.

Theorem 3.1.7 (Linear Properties of Laplace Transform)

$$\mathcal{L}[f(t) + g(t)] = \mathcal{L}[f(t)] + \mathcal{L}[g(t)]$$

$$\mathcal{L}[kf(t)] = k \mathcal{L}[f(t)] \text{ for any constant } k$$

Example 3.1.8

$$(a) \quad \mathcal{L}[2e^{-t} + \sin 3t] = 2\mathcal{L}[e^{-t}] + \mathcal{L}[\sin 3t] = \frac{2}{s - (-1)} + \frac{3}{s^2 + 3^2} = \frac{2}{s + 1} + \frac{3}{s^2 + 9}$$

$$(b) \quad \mathcal{L}[1 + \delta(t)] = \mathcal{L}[e^0] + \mathcal{L}[\delta(t)] = \frac{1}{s - 0} + e^{-0s} = \frac{1}{s} + 1$$

$$(c) \quad \mathcal{L}^{-1}\left[e^{-s} + 1 + \frac{1}{s}\right] = \mathcal{L}^{-1}[e^{-s}] + \mathcal{L}^{-1}[e^{-0s}] + \mathcal{L}^{-1}\left[\frac{1}{s}\right] = \delta(t - 1) + \delta(t) + 1$$

$$(d) \quad \mathcal{L}^{-1}\left[\frac{s + 3}{s^2 + 9}\right] = \mathcal{L}^{-1}\left[\frac{s}{s^2 + 3^2}\right] + \mathcal{L}^{-1}\left[\frac{3}{s^2 + 3^2}\right] = \cos 3t + \sin 3t$$

Given two functions $f(t)$ and $g(t)$, the **convolution** $(f * g)(t)$ is defined by

$$(f * g)(t) = \int_0^t f(u)g(t - u)du$$

The Laplace transform of the convolution of f and g is the product of the Laplace transform of f and the Laplace transform of g , that is,

$$\mathcal{L}[(f * g)(t)] = \mathcal{L}[f(t)] \cdot \mathcal{L}[g(t)].$$

In other words, $\mathcal{L}[f * g] = F(s)G(s)$.

Example 3.1.9

$$(a) \quad \mathcal{L}[t * \sin t] = \mathcal{L}[t] \cdot \mathcal{L}[\sin t] = \frac{1!}{s^{1+1}} \cdot \frac{1}{s^2 + 1^2} = \frac{1}{s^2(s^2 + 1)}$$

$$(b) \quad \mathcal{L}[t * e^t] = \mathcal{L}[t] \cdot \mathcal{L}[e^t] = \frac{1}{s^2} \cdot \frac{1}{s - 1} = \frac{1}{s^2(s - 1)}$$

$$(c) \quad \mathcal{L}[e^t * \cos t] = \mathcal{L}[e^t] \cdot \mathcal{L}[\cos t] = \frac{1}{s - 1} \cdot \frac{s}{s^2 + 1} = \frac{s}{(s - 1)(s^2 + 1)}$$

Note that the symbol $*$ is used for convolution, not multiplication. Hence, the expressions $t * \sin t$ and $t \sin t$ are completely different. To find the Laplace transform of $t \sin t$, or in general, $t f(t)$, we apply the formula

$$\mathcal{L}[t f(t)] = -\frac{d}{ds}(\mathcal{L}[f(t)]).$$

Alternatively, we write $\mathcal{L}[t f(t)] = -F'(s)$.

Example 3.1.10

Evaluate $\mathcal{L}[t \sin t]$.

Solution

$$\text{Since } \mathcal{L}[\sin t] = \frac{1}{s^2 + 1},$$

$$\begin{aligned} \mathcal{L}[t \sin t] &= -\frac{d}{ds} \left(\frac{1}{s^2 + 1} \right) = -\frac{d}{ds} ((s^2 + 1)^{-1}) \\ &= (s^2 + 1)^{-2} (2s) \\ &= \frac{2s}{(s^2 + 1)^2} \end{aligned}$$

3.2 Laplace Transforms of Products of e^{at}

The following theorem considers the Laplace transform of a product involving exponential function e^{at} . It is also known as the s -shift theorem or First Shift Theorem.

Theorem 3.2.1

If $F(s) = \mathcal{L}[f(t)]$, then

$$\mathcal{L}[e^{at}f(t)] = F(s - a).$$

Proof

From Definition 3.1.1, we have

$$\begin{aligned}\mathcal{L}[e^{at}f(t)] &= \lim_{R \rightarrow \infty} \int_0^R e^{-st} \cdot e^{at}f(t)dt \\ &= \lim_{R \rightarrow \infty} \int_0^R e^{-(s-a)t}f(t)dt \\ &= F(s - a).\end{aligned}$$

Applying Theorem 3.2.1 for Laplace Transform

Step 1. Identify a and $f(t)$.

Step 2. Evaluate $F(s) = \mathcal{L}[f(t)]$.

Step 3. Conclude $\mathcal{L}[e^{at}f(t)] = F(s - a)$.

Example 3.2.2Evaluate $\mathcal{L}[e^t \sin t]$.**Solution**Step 1. $a = 1$ and $f(t) = \sin t$ Step 2. $F(s) = \mathcal{L}[\sin t] = \frac{1}{s^2 + 1}$ Step 3. $\mathcal{L}[e^t \sin t] = F(s - 1) = \frac{1}{(s - 1)^2 + 1}$ **Example 3.2.3**Evaluate $\mathcal{L}[t^2 e^{-t}]$.**Solution**Step 1. $a = -1$ and $f(t) = t^2$ Step 2. $F(s) = \mathcal{L}[t^2] = \frac{2!}{s^{2+1}} = \frac{2}{s^3}$ Step 3. $\mathcal{L}[t^2 e^{-t}] = F(s + 1) = \frac{2}{(s + 1)^3}$ **Example 3.2.4**Evaluate $\mathcal{L}[e^{2t} \delta(t - 1)]$.**Solution**Step 1. $a = 2$ and $f(t) = \delta(t - 1)$ Step 2. $F(s) = \mathcal{L}[\delta(t - 1)] = e^{-s}$ Step 3. $\mathcal{L}[e^{2t} \delta(t - 1)] = F(s - 2) = e^{-(s-2)}$

Theorem 3.2.1 for inverse Laplace transform is

$$\mathcal{L}^{-1}[F(s - a)] = e^{at} f(t).$$

Applying Theorem 3.2.1 for Inverse Laplace Transform

Step 1. Identify a and find $F(s)$ from $F(s - a)$.

Step 2. Evaluate $f(t) = \mathcal{L}^{-1}[F(s)]$.

Step 3. Conclude $\mathcal{L}^{-1}[F(s - a)] = e^{at} f(t)$.

Remark. To identify a , we observe the factor $(s - a)$ in the denominator, e.g. $(s - 1)^2$, $(s + 2)$, $(s - 2)^3$, etc.

Example 3.2.5

Evaluate $\mathcal{L}^{-1}\left[\frac{1}{(s - 1)^2 + 1}\right]$.

Solution

Step 1. $a = 1$ and $F(s - 1) = \frac{1}{(s - 1)^2 + 1}$. So, $F(s) = \frac{1}{s^2 + 1}$

Step 2. $f(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2 + 1}\right] = \sin t$

Step 3. $\mathcal{L}^{-1}\left[\frac{1}{(s - 1)^2 + 1}\right] = e^t \sin t$

Example 3.2.6

Evaluate $\mathcal{L}^{-1}\left[\frac{2}{(s+1)^3}\right]$.

Solution

Step 1. $a = -1$ and $F(s+1) = \frac{2}{(s+1)^3}$. So, $F(s) = \frac{2}{s^3}$

Step 2. $f(t) = \mathcal{L}^{-1}\left[\frac{2}{s^3}\right] = \mathcal{L}^{-1}\left[\frac{2!}{s^{2+1}}\right] = t^2$

Step 3. $\mathcal{L}^{-1}\left[\frac{2}{(s+1)^3}\right] = e^{-t} t^2$

Example 3.2.7

Evaluate $\mathcal{L}^{-1}[e^{2-s}]$.

Solution

Step 1. $e^{2-s} = e^{-(s-2)} \Rightarrow a = 2$ and $F(s-2) = e^{-(s-2)}$. So, $F(s) = e^{-s}$

Step 2. $f(t) = \mathcal{L}^{-1}[e^{-s}] = \delta(t-1)$

Step 3. $\mathcal{L}^{-1}[e^{2-s}] = e^{2t} \delta(t-1)$

An alternative answer for $\mathcal{L}^{-1}[e^{2-s}]$ is

$$\mathcal{L}^{-1}[e^{2-s}] = e^2 \mathcal{L}^{-1}[e^{-s}] = e^2 \delta(t-1).$$

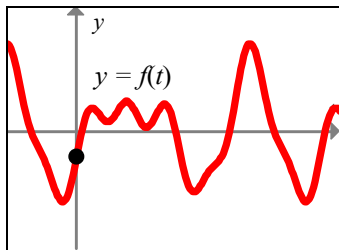
The functions $e^{2t} \delta(t-1)$ and $e^2 \delta(t-1)$ are equivalent because of the property of the delta function.

3.3 Laplace Transform of a Truncated Shift of a Function

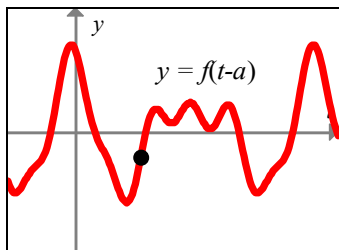
For a given function $f(t)$ and positive constant a , the truncated shift of f by a units is given by

$$f(t - a) U(t - a)$$

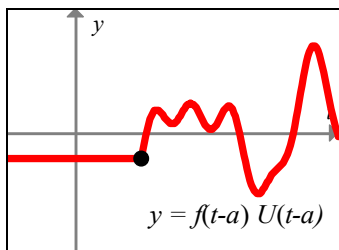
Figure 3.5 below compares the graphs of $f(t)$, $f(t - a)$, and $f(t - a)U(t - a)$.



Original signal $y = f(t)$



$y = f(t - a)$ is the shift of $y = f(t)$ by a units to the right



$y = f(t - a)U(t - a)$ is the truncated shift of $y = f(t)$

Figure 3.5. Graphs of $f(t)$, $f(t - a)$, and $f(t - a)U(t - a)$

Theorem 3.3.1 considers the Laplace transform of a truncated shift of a function. It is also known as the Second Shift Theorem or t -shift Theorem.

Theorem 3.3.1

If $F(s) = \mathcal{L}[f(t)]$, then

$$\mathcal{L}[f(t - a) U(t - a)] = e^{-as} F(s)$$

Proof

From Definition 3.1.1, we have

$$\begin{aligned} \mathcal{L}[f(t - a) U(t - a)] &= \lim_{R \rightarrow \infty} \int_0^R e^{-st} \cdot f(t - a) U(t - a) dt \\ &= \lim_{R \rightarrow \infty} \int_a^R e^{-st} f(t - a) dt = \lim_{R \rightarrow \infty} \int_0^R e^{-s(u+a)} f(u) du \\ &= e^{-as} \lim_{R \rightarrow \infty} \int_0^R e^{-su} f(u) du = e^{-as} F(s) \end{aligned}$$

Applying Theorem 3.3.1 for Laplace Transform

Step 1. Identify a and find $f(t)$ from $f(t - a)$.

Step 2. Evaluate $F(s) = \mathcal{L}[f(t)]$.

Step 3. Conclude $\mathcal{L}[f(t - a) U(t - a)] = e^{-as} F(s)$.

Remark. To identify a , we observe the factor $U(t - a)$.

Example 3.3.2Evaluate $\mathcal{L}[t U(t - 1)]$.**Solution**Step 1. $a = 1$ and $f(t - 1) = t$. So, $f(t) = t + 1$ Step 2. $F(s) = \mathcal{L}[t + 1] = \frac{1}{s^2} + \frac{1}{s}$ Step 3. $\mathcal{L}[t U(t - 1)] = e^{-s} \left(\frac{1}{s^2} + \frac{1}{s} \right)$ **Example 3.3.3**Evaluate $\mathcal{L}[\sin t U(t - \pi)]$.**Solution**Step 1. $a = \pi$ and $f(t - \pi) = \sin t$. So, $f(t) = \sin(t + \pi) = -\sin t$ Step 2. $F(s) = \mathcal{L}[-\sin t] = \frac{-1}{s^2 + 1}$ Step 3. $\mathcal{L}[\sin t U(t - \pi)] = e^{-\pi s} \left(\frac{-1}{s^2 + 1} \right)$ **Example 3.3.4**Evaluate $\mathcal{L}[(t^2 - 1) U(t - 1)]$.**Solution**Step 1. $a = 1$ and $f(t - 1) = t^2 - 1$. So, $f(t) = (t + 1)^2 - 1 = t^2 + 2t$ Step 2. $F(s) = \mathcal{L}[t^2 + 2t] = \frac{2}{s^3} + \frac{2}{s^2}$ Step 3. $\mathcal{L}[(t^2 - 1) U(t - 1)] = e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} \right)$

Theorem 3.3.1 for inverse Laplace transform is

$$\mathcal{L}^{-1}[e^{-as}F(s)] = f(t - a) U(t - a).$$

Applying Theorem 3.3.1 for Inverse Laplace Transform

Step 1. Identify a and $F(s)$.

Step 2. Evaluate $f(t) = \mathcal{L}^{-1}[F(s)]$.

Step 3. Find $f(t - a)$ and conclude $\mathcal{L}^{-1}[e^{-as}F(s)] = f(t - a)U(t - a)$.

Remark. We observe the exponential factor e^{-as} to identify the value of a .

Example 3.3.5

Evaluate $\mathcal{L}^{-1}\left[e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right)\right]$.

Solution

Step 1. $a = 1$ and $F(s) = \frac{1}{s^2} + \frac{1}{s}$

Step 2. $f(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2} + \frac{1}{s}\right] = t + 1$

Step 3. $f(t - 1) = (t - 1) + 1 = t$ and

$$\mathcal{L}^{-1}\left[e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right)\right] = t U(t - 1)$$

Example 3.3.6

Evaluate $\mathcal{L}^{-1} \left[e^{-\pi s} \left(\frac{-1}{s^2 + 1} \right) \right]$.

Solution

Step 1. $a = \pi$ and $F(s) = \frac{-1}{s^2 + 1}$

Step 2. $f(t) = \mathcal{L}^{-1} \left[\frac{-1}{s^2 + 1} \right] = -\sin t$

Step 3. $f(t - \pi) = -\sin(t - \pi) = \sin t$ and

$$\mathcal{L}^{-1} \left[e^{-\pi s} \left(\frac{-1}{s^2 + 1} \right) \right] = \sin t \, U(t - \pi)$$

Example 3.3.7

Evaluate $\mathcal{L}^{-1} \left[e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} \right) \right]$.

Solution

Step 1. $a = 1$ and $F(s) = \frac{2}{s^3} + \frac{2}{s^2}$

Step 2. $f(t) = \mathcal{L}^{-1} \left[\frac{2}{s^3} + \frac{2}{s^2} \right] = \mathcal{L}^{-1} \left[\frac{2!}{s^{2+1}} \right] + 2\mathcal{L}^{-1} \left[\frac{1!}{s^{1+1}} \right] = t^2 + 2t$

Step 3. $f(t - 1) = (t - 1)^2 + 2(t - 1) = t^2 - 1$ and

$$\mathcal{L}^{-1} \left[e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} \right) \right] = (t^2 - 1) U(t - 1)$$

3.4 Application

The Laplace transform and its inverse is useful for solving initial value problems (IVP) involving linear differential equations. This algebraic technique provides an alternative to those in Chapters 1 and 2. It is particularly useful when dealing with the unit-step and delta functions. Figure 3.6 below illustrates the concept of the Laplace transform (and its inverse) in solving IVPs.



Figure 3.6. The Laplace transform technique for solving IVPs

The Laplace transform technique allows us to find the particular solution of higher order linear ODEs efficiently. To use this technique, we need the Laplace transform of the first and second order derivatives of the function f .

$$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0)$$

$$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0) - f'(0)$$

Procedure to Solve IVP using Laplace Transform

- Step 1. Apply Laplace transforms on both sides of the differential equation.
- Step 2. Substitute initial values $f(0)$ and $f'(0)$ and express $F(s)$ as the subject.
- Step 3. Evaluate $f(t) = \mathcal{L}^{-1}[F(s)]$.

Compare the technique used in Example 3.4.1 below with that in Example 1.4.6 of Chapter 1.

Example 3.4.1

Solve the IVP:

$$\begin{cases} \frac{df}{dt} + f = 0 \\ f(0) = 10 \end{cases}$$

Solution

$$\mathcal{L}\left[\frac{df}{dt}\right] + \mathcal{L}[f] = \mathcal{L}[0] \Rightarrow (s F(s) - f(0)) + F(s) = 0$$

$$\Rightarrow (s F(s) - 10) + F(s) = 0$$

$$\Rightarrow (s + 1) F(s) = 10 \Rightarrow F(s) = \frac{10}{s + 1}$$

$$f(t) = \mathcal{L}^{-1}\left[\frac{10}{s+1}\right] = 10 e^{-t}$$

In many cases, it is necessary to evaluate the inverse Laplace transform (in the last step above) through Partial Fraction Decomposition (PFD), see Example 3.4.2 below.

Example 3.4.2

Use PFD to evaluate $\mathcal{L}^{-1}\left[\frac{1}{s(s^2 + 1)}\right]$.

Solution

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1} \Rightarrow 1 = A(s^2 + 1) + (Bs + C)s$$

Using coefficients comparison (or other method) we have $A = 1$, $B = -1$, $C = 0$.

$$\mathcal{L}^{-1}\left[\frac{1}{s(s^2 + 1)}\right] = \mathcal{L}^{-1}\left[\frac{1}{s} + \frac{-s + 0}{s^2 + 1}\right] = \mathcal{L}^{-1}\left[\frac{1}{s}\right] - \mathcal{L}^{-1}\left[\frac{s}{s^2 + 1}\right] = 1 - \cos t$$

Example 3.4.3

Solve the IVP:

$$\begin{cases} \frac{d^2 f}{dt^2} + f = 1 \\ f(0) = 0 \\ f'(0) = 0 \end{cases}$$

Solution

$$\mathcal{L}\left[\frac{d^2 f}{dt^2}\right] + \mathcal{L}[f] = \mathcal{L}[1]$$

$$\Rightarrow (s^2 F(s) - sf(0) - f'(0)) + F(s) = \frac{1}{s}$$

$$\Rightarrow (s^2 F(s) - 0 - 0) + F(s) = \frac{1}{s}$$

$$\Rightarrow (s^2 + 1)F(s) = \frac{1}{s}$$

$$\Rightarrow F(s) = \frac{1}{s(s^2 + 1)}$$

From Example 3.4.2, we have

$$f(t) = \mathcal{L}^{-1}\left[\frac{1}{s(s^2 + 1)}\right] = 1 - \cos t$$

Example 3.4.4

Solve the IVP:

$$\begin{cases} \frac{d^2 f}{dt^2} = 2U(t-1) \\ f(0) = 0 \\ f'(0) = 0 \end{cases}$$

Solution

$$\mathcal{L}\left[\frac{d^2 f}{dt^2}\right] = 2\mathcal{L}[U(t-1)]$$

$$\Rightarrow s^2 F(s) - sf(0) - f'(0) = 2 \frac{e^{-s}}{s}$$

$$\Rightarrow s^2 F(s) - 0 - 0 = 2 \frac{e^{-s}}{s}$$

$$\Rightarrow s^2 F(s) = 2 \frac{e^{-s}}{s} \Rightarrow F(s) = 2 \frac{e^{-s}}{s^3}$$

$$f(t) = \mathcal{L}^{-1}\left[e^{-s}\left(\frac{2}{s^3}\right)\right] = g(t-1)U(t-1)$$

$$G(s) = \frac{2}{s^3} \Rightarrow g(t) = t^2 \Rightarrow g(t-1) = (t-1)^2$$

$$\text{Thus, } f(t) = (t-1)^2 U(t-1).$$

Example 3.4.5

A series circuit consists of an inductor of inductance 1H, a resistor of resistance 0.4 k Ω , and a capacitor of capacitance 25 μ F. An impulse is applied onto the closed circuit so that the charge q (in coulomb) on the capacitor satisfies the differential equation

$$\frac{d^2q}{dt^2} + 400 \frac{dq}{dt} + 40000q = 100\delta(t - 5)$$

Assuming zero initial conditions, express q in terms of t .

Solution

$$\mathcal{L}\left[\frac{d^2q}{dt^2}\right] + 400\mathcal{L}\left[\frac{dq}{dt}\right] + 40000\mathcal{L}[q] = 100\mathcal{L}[\delta(t - 5)]$$

$$\Rightarrow (s^2Q(s) - sq(0) - q'(0)) + 400(sQ(s) - q(0)) + 40000Q(s) = 100e^{-5s}$$

$$\Rightarrow s^2Q(s) + 400sQ(s) + 40000Q(s) = 100e^{-5s}$$

$$\Rightarrow (s^2 + 400s + 40000)Q(s) = 100e^{-5s}$$

$$\Rightarrow Q(s) = \frac{100e^{-5s}}{s^2 + 400s + 40000} = e^{-5s} \left(\frac{100}{(s + 200)^2} \right)$$

$$q(t) = \mathcal{L}^{-1} \left[e^{-5s} \left(\frac{100}{(s + 200)^2} \right) \right] = f(t - 5)U(t - 5)$$

$$F(s) = \frac{100}{(s + 200)^2} \Rightarrow f(t) = \mathcal{L}^{-1} \left[\frac{100}{(s + 200)^2} \right] = e^{-200t}g(t)$$

$$G(s + 200) = \frac{100}{(s + 200)^2} \Rightarrow G(s) = \frac{100}{s^2} \Rightarrow g(t) = 100t$$

$$f(t) = \mathcal{L}^{-1} \left[\frac{100}{(s + 200)^2} \right] = e^{-200t}(100t)$$

$$\text{Thus, } q(t) = f(t - 5)U(t - 5) = 100e^{-200(t-5)}(t - 5)U(t - 5).$$

3.5 Exercises

Part A: Baseline Problems

1. Evaluate the following Laplace transforms (see Examples 3.1.2 and 3.1.3).

(a) $\mathcal{L}[1 + t]$

(c) $\mathcal{L}[2t^2 + e^{2t}]$

(b) $\mathcal{L}[t^2 + 2t - 1]$

(d) $\mathcal{L}[e^{-2t} - e^{3t}]$

2. Evaluate the following inverse Laplace transforms (see Examples 3.1.2 and 3.1.3).

(a) $\mathcal{L}^{-1}\left[\frac{2}{s^3} + \frac{1}{s^2}\right]$

(c) $\mathcal{L}^{-1}\left[\frac{4}{s^5} + \frac{3}{s^4} - \frac{1}{s+5}\right]$

(b) $\mathcal{L}^{-1}\left[\frac{1}{s} - \frac{1}{s-4}\right]$

(d) $\mathcal{L}^{-1}\left[\frac{2}{s+1} - \frac{1}{s-2}\right]$

3. Evaluate the following Laplace transforms (see Example 3.1.4).

(a) $\mathcal{L}[\sin 3t]$

(c) $\mathcal{L}[\cos 4t + 3 \sin 2t]$

(b) $\mathcal{L}[\sin t - \cos t]$

(d) $\mathcal{L}[\sin 5t - \sin t + \cos 2t]$

4. Evaluate the following inverse Laplace transforms (see Examples 3.1.3 and 3.1.4).

(a) $\mathcal{L}^{-1}\left[\frac{2}{s^2+1} + \frac{2}{s+1}\right]$

(c) $\mathcal{L}^{-1}\left[\frac{s+3}{s^2+9}\right]$

(b) $\mathcal{L}^{-1}\left[\frac{1}{s+4} + \frac{1}{s^2+4}\right]$

(d) $\mathcal{L}^{-1}\left[\frac{2-3s}{s^2+25}\right]$

5. Evaluate the following Laplace transforms (see Example 3.1.5).

(a) $\mathcal{L}[U(t-1) - U(t-3)]$ (c) $\mathcal{L}[3\delta(t-2)]$

(b) $\mathcal{L}[2U(t-3)]$ (d) $\mathcal{L}[5U(t-2) + \delta(t-4)]$

6. Evaluate the following inverse Laplace transforms (see Example 3.1.5).

(a) $\mathcal{L}^{-1}[e^{-2s}]$ (c) $\mathcal{L}^{-1}\left[2e^{-3s} + 3\frac{e^{-3s}}{s}\right]$

(b) $\mathcal{L}^{-1}\left[\frac{e^{-2s}}{s}\right]$ (d) $\mathcal{L}^{-1}\left[e^{-s} + \frac{2e^{-2s}}{s}\right]$

7. Evaluate the following Laplace transforms (see Example 3.1.9).

(a) $\mathcal{L}[t * \sin 4t]$

(b) $\mathcal{L}[t^2 * \cos 2t]$

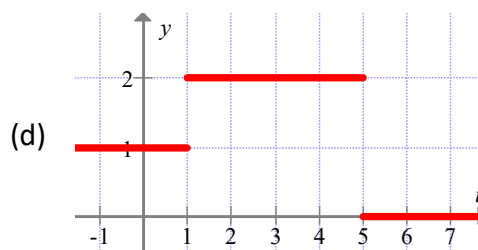
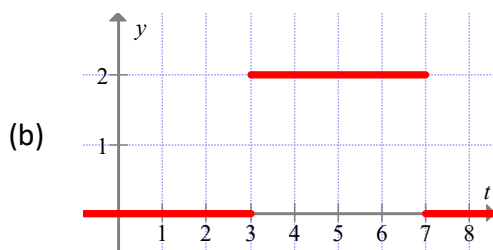
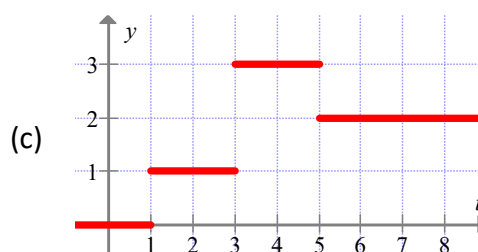
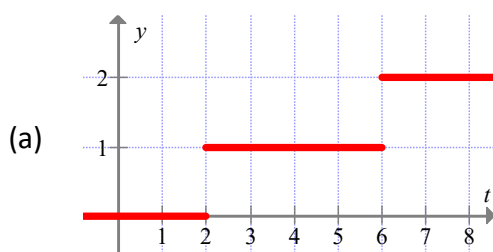
(c) $\mathcal{L}[t * e^{-t}]$

(d) $\mathcal{L}[e^{2t} * U(t-3)]$

(e) $\mathcal{L}[t * e^{2t} * \sin t]$

Part B: Intermediate Problems

8. The graphs of $f(t)$ is given below. Express $f(t)$ in terms of unit-step functions, and evaluate $\mathcal{L}[f(t)]$ (see Example 3.1.6).



9. Evaluate the following Laplace transforms (see Examples 3.2.2 – 3.2.4).

(a) $\mathcal{L}[e^t \sin 2t]$

(c) $\mathcal{L}[t e^{-2t}]$

(b) $\mathcal{L}[e^{-t} \cos t]$

(d) $\mathcal{L}[t^3 e^{3t}]$

10. Evaluate the following inverse Laplace transforms (see Examples 3.2.5 – 3.2.7).

(a) $\mathcal{L}^{-1}\left[\frac{1}{(s-2)^2 + 1}\right]$

(c) $\mathcal{L}^{-1}\left[\frac{s}{(s-1)^2 + 4}\right]$

(b) $\mathcal{L}^{-1}\left[\frac{2}{(s-1)^3}\right]$

(d) $\mathcal{L}^{-1}\left[\frac{s-2}{(s+2)^2 + 4}\right]$

11. Evaluate the following Laplace transforms (see Examples 3.3.2 – 3.3.4).

(a) $\mathcal{L}[t^2 U(t - 2)]$ (c) $\mathcal{L}[(2t + 1) U(t - 5)]$

(b) $\mathcal{L}\left[\sin t U\left(t - \frac{\pi}{2}\right)\right]$ (d) $\mathcal{L}[(t^3 - 1) U(t - 1)]$

12. Evaluate the following inverse Laplace transforms (see Examples 3.3.5 – 3.3.7).

(a) $\mathcal{L}^{-1}\left[\frac{e^{-s}}{s^3}\right]$ (c) $\mathcal{L}^{-1}\left[e^{-\pi s}\left(\frac{s + 3}{s^2 + 9}\right)\right]$

(b) $\mathcal{L}^{-1}\left[e^{-2s}\left(\frac{2}{s^3} + \frac{1}{s^2}\right)\right]$ (d) $\mathcal{L}^{-1}\left[e^{-2\pi s}\left(\frac{2s - 5}{s^2 + 4}\right)\right]$

13. Evaluate the following Laplace transforms (see Example 3.1.10).

(a) $\mathcal{L}[t \sin 2t]$ (c) $\mathcal{L}[t \cos 2t]$

(b) $\mathcal{L}[t \cos t]$ (d) $\mathcal{L}[t^2 \sin t]$

14. Evaluate the following Laplace transforms.

(a) $\mathcal{L}[t e^{2t} U(t - 1)]$ (c) $\mathcal{L}[t^2 e^{-t} U(t - 2)]$

(b) $\mathcal{L}\left[e^t \sin t U\left(t - \frac{\pi}{2}\right)\right]$ (d) $\mathcal{L}\left[e^{-2t} \cos 2t U\left(t - \frac{\pi}{2}\right)\right]$

15. Evaluate the following inverse Laplace transforms (see Example 3.4.2).

(a) $\mathcal{L}^{-1}\left[\frac{s - 1}{s(s^2 + 1)}\right]$ (c) $\mathcal{L}^{-1}\left[\frac{3}{s^2(s - 3)}\right]$

(b) $\mathcal{L}^{-1}\left[\frac{s}{s^2 - 1}\right]$ (d) $\mathcal{L}^{-1}\left[\frac{6}{(s^2 + 1)(s^2 + 4)}\right]$

16. Evaluate the following inverse Laplace transforms.

$$(a) \quad \mathcal{L}^{-1} \left[e^{-4s} \frac{s-1}{s(s^2+1)} \right]$$

$$(c) \quad \mathcal{L}^{-1} \left[e^{-2s} \frac{2}{s^2(s-3)} \right]$$

$$(b) \quad \mathcal{L}^{-1} \left[e^{-s} \frac{s}{s^2-1} \right]$$

$$(d) \quad \mathcal{L}^{-1} \left[e^{-3s} \frac{6}{(s^2+1)(s^2+4)} \right]$$

Part C: Application Problems

(See Section 3.4)

17. Solve the following IVPs using Laplace transform (see Examples 3.4.3 – 3.4.4).

$$(a) \quad \begin{cases} \frac{df}{dt} = U(t-1) \\ f(0) = 3 \end{cases}$$

$$(c) \quad \begin{cases} \frac{df}{dt} = t U(t-2) \\ f(0) = 4 \end{cases}$$

$$(b) \quad \begin{cases} \frac{d^2f}{dt^2} + f = 1 - U(t-1) \\ f(0) = 0 \\ f'(0) = 0 \end{cases}$$

$$(d) \quad \begin{cases} \frac{d^2f}{dt^2} + f = \delta(t-1) \\ f(0) = 0 \\ f'(0) = 2 \end{cases}$$

18. A series circuit consists of an inductor of inductance 2H, a resistor of resistance 0.1 k Ω , and a capacitor of capacitance 0.8 mF. An impulse is applied onto the closed circuit so that the charge q (in coulomb) on the capacitor satisfies the differential equation

$$\frac{d^2q}{dt^2} + 50 \frac{dq}{dt} + 625q = 110 \delta(t-2)$$

Assuming zero initial conditions, express q in terms of t .

19. A series circuit consists of a capacitor of capacitance 10 mF and an inductor of inductance 1H. A timer is applied onto the closed circuit so that the charge q (in coulomb) on the capacitor satisfies the differential equation

$$\frac{d^2q}{dt^2} + 100q = 12 U(t - \pi)$$

Assuming zero initial conditions, express q in terms of t .

20. A spring of stiffness 10 N/m lies on a smooth horizontal surface with a smooth ball of unit mass is attached to its end, while the other end is fixed to a wall. Let x be the displacement of the ball from its equilibrium position. An impulse force acts on the ball so that its harmonic motion satisfies the differential equation

$$\frac{d^2x}{dt^2} + 4x = \delta(t - \pi).$$

Express the displacement x as a function of t .